# Full Stack Software Engineer Thought Exercise

[February 2026]

**Overview**
E3n is building and evolving a modern data platform that powers our achievement and admissions products. In this role, it is critical that a Full Stack Engineer can design clean, scalable systems; reason about data persistence; and write maintainable code with clear documentation. The exercise below is intended to assess your ability to design and implement a small system end-to-end (including data modeling) and to communicate your technical decisions clearly.

**Your Assignment: Game of War**
War is a card game involving two players. The deck is divided evenly among the two players, giving each a down stack. In unison, each player reveals the top card on his deck (a "battle"), and the player with the higher card takes both the cards played and moves them to the bottom of their stack. If the two cards played are of equal value, each player lays down three face-down cards and picks one of the cards out of the three (a "war"), and the higher-valued card wins all of the cards on the table, which are then added to the bottom of the player's stack. If one of the players has no more cards in a battle that player loses that battle. In the case of another tie, the war process is repeated until there is no tie.

A player wins by collecting all the cards. If a player runs out of cards while dealing the face-down cards of a war, they may play the last card in his deck as their face-up card and still have a chance to stay in the game.

Given these rules, define the classes you would implement to build a War simulator, which would log results for each game and each round of each game in a database. The database must also catalog the winner of each round and the winner of each game.

**Response Requirements & Time Limit**
Please implement using a language/framework you are comfortable with (ideally aligned to the JD stack). In addition your submission should include:

1. Your SQL schema (DDL) and any brief notes/assumptions.
2. Working code for your simulator.
3. Brief documentation explaining your design decisions, how to run the code, and how results are persisted (including how each round/game winner is stored).

**Submission format:** Please share a link to a Git repository (preferred). If that is not possible, you may submit a zipped folder containing your code and documentation.

**Time expectation:** This exercise is designed to take approximately **4–6 hours**. Please time-box your effort and include a short note describing what you would improve or add with additional time.

**Timing:** Please submit your materials within **72 hours** of receiving the exercise.

**Exercise Submission**
Please email your response to Karen with "(Your Name) - E3n Full Stack Software Engineer Thought Exercise Response" in the subject line of your email. Should you have any questions regarding this exercise, please direct your questions to Karen at kclark@erblearn.org.

**AI Usage Disclosure**
To ensure we accurately assess your individual technical skills and problem-solving approach, the use of generative AI tools (e.g., ChatGPT, Copilot, Claude, etc.) to generate or substantially assist with code, architecture design, or written responses for this exercise is not permitted.

By submitting your solution, you confirm that the work represents your own original thinking and effort.